

OpenPI Comet

Empiricist · seat 4

H1 horizon ablation · setup/run · GPU budget

11+3 min

arXiv 2512.10071 · BEHAVIOR Challenge 2025 · smoke-run paper club

H1 statement

H1 — Action horizon under fixed recipe

RH + abs joints + proprio + RGB · task: turning_on_radio

H = 8

SR 0.00

paper Table 3 #2



H = 32

SR 0.30

>> (non-monotonic)

Horizon table — Table 3 #2

Action horizon ablation (Table 3 #2)

Receding Horizon · RGB · 224×224 · turning_on_radio



Sweet spot H=32 — too short fails; too long conflicts

Table 3 excerpt (paper)

rate tasks are not shown.

Table 3: Ablation study of control mode, action horizon, input modality, and image resolution on the turning on radio task.

Settings	Control Mode	Action Horizon	Input Modality	Image Resolution	Success Rate
#1	Temporal Ensemble	50	RGB	Head:224	0.00
	Receding Temporal			Wrist:224	0.00
	Receding Horizon				0.25
#2	Receding Horizon	8	RGB	Head:224 Wrist:224	0.00
		16			0.10
		50			0.25
		32			0.30
#3	Receding Horizon	32	RGB+Depth Image	Head:224	0.20
			RGB+Point Cloud	Wrist:224	0.30
			RGB		0.30
#4	Receding Horizon	32	RGB	Head:224 Wrist:224	0.30
				Head:720 Wrist:480	0.60

Implement steps

Implement — smoke E01 path

1

Clone

openpi-comet + BEHAVIOR-1K

2

Warm-start

HF pi05-b1kpt12-cs32

3

Two SFT

radio_h32 / radio_h8

~2k steps

4

Serve

serve_b1k

receeding_horizon

5

Eval

OmniGibson

5-10 instances

GPU budget

GPU budget — E01 smoke (~15-50 GPU-h)

SFT H=32 · ~2k steps

~2-6 GPU-h

LoRA >22.5 GB / FT >70 GB

SFT H=8 · same

~2-6 GPU-h

same

Eval 5-10 × 2 ckpts

~10-40 GPU-h

sim-bound; infer >8 GB

TOTAL

~15-50 GPU-h

Prefer 1×A100-80 / H100 / H200

Success criteria (smoke)

IN SCOPE

Ordinal gap $H_{32}-H_8 \geq +0.15$
(or $H_{32}>0$ and $H_8\approx 0$)

OUT OF SCOPE

Absolute SR 0.30
Full challenge Q / 50-task

OPTIONAL E02

Fix H32 ckpt; eval-only
RH vs temporal ensemble

**Falsify the horizon claim on one easy task
first — then scale.**

Q&A · timing 11+3